

جامعة أم القرى
UMM AL-QURA UNIVERSITY

# Vertebral Column(The Neck)

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# Lecture Objectives

**Learning Objectives**

Detailed anatomical organisation of the upper (cervical) vertebral column - namely the neck; the bones, joints, muscles (movements) and nerves

**Aim**

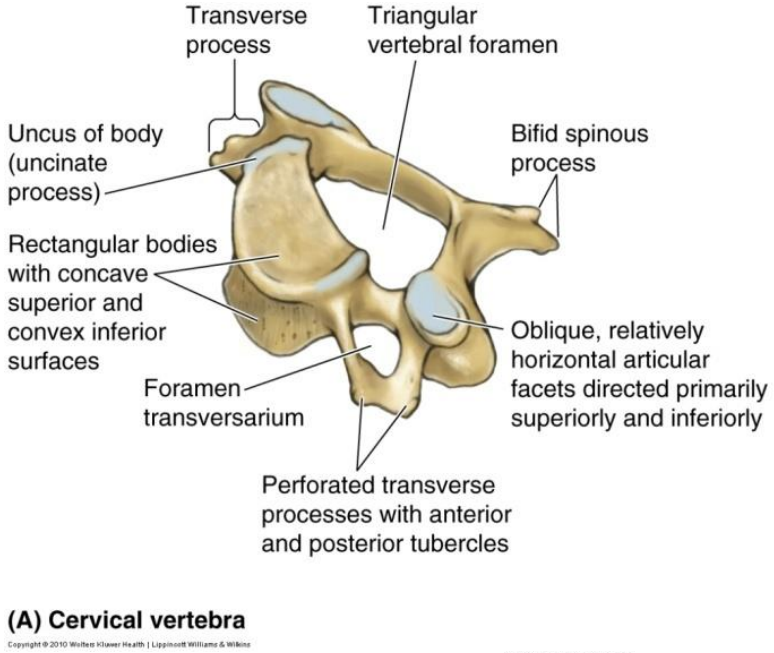
To understand the general structure of the cervical part of the vertebral column

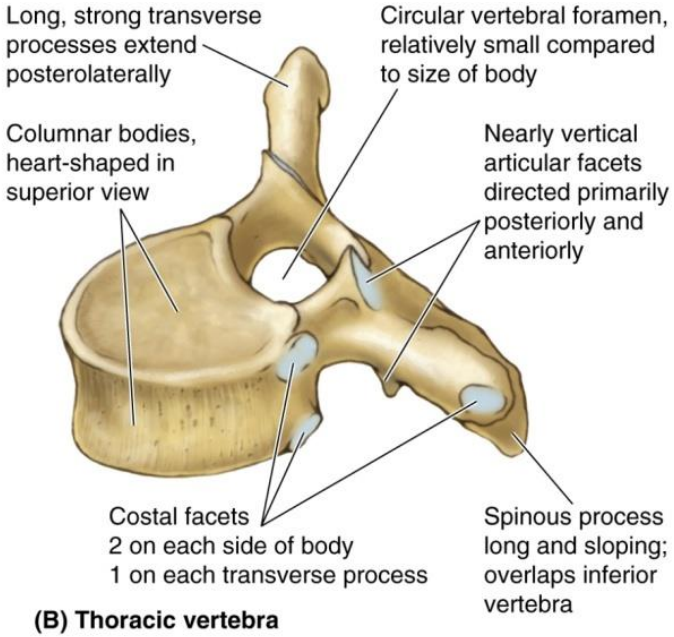
**Content**

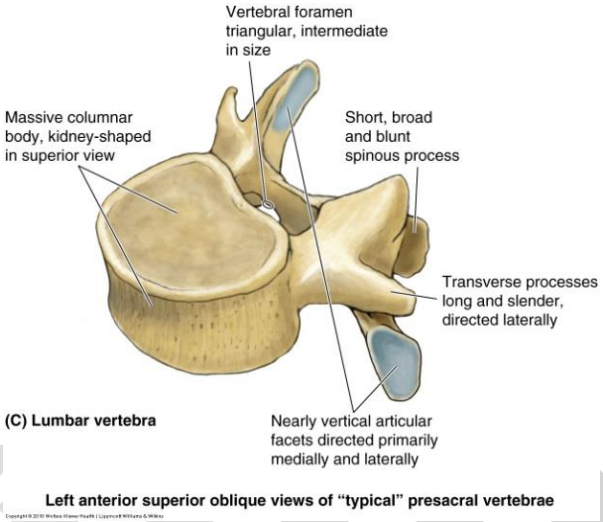
- To identify **cervical vertebrae** and their parts on isolated bones, articulated skeletons and radiographs
- To name and describe the **joints of the cervical spine** and state the movements that take place at these joints
- To describe the boundaries and **contents of the cervical vertebral canal** and the effect of impingement on the spinal cord in this location.
- To describe the exit points of cervical spinal nerves and common causes of **IVF encroachment**
- To know the level of the nerve roots impinged upon at a given IVF in cervical spine e.g. C5/6 IVF
- To describe and demonstrate the **dermatome and myotome tests** for the upper limb
- To describe the course of the **vertebral artery** and the parts of the nervous system it supplies

# Typical Cervical Vertebrae (C3-C7)

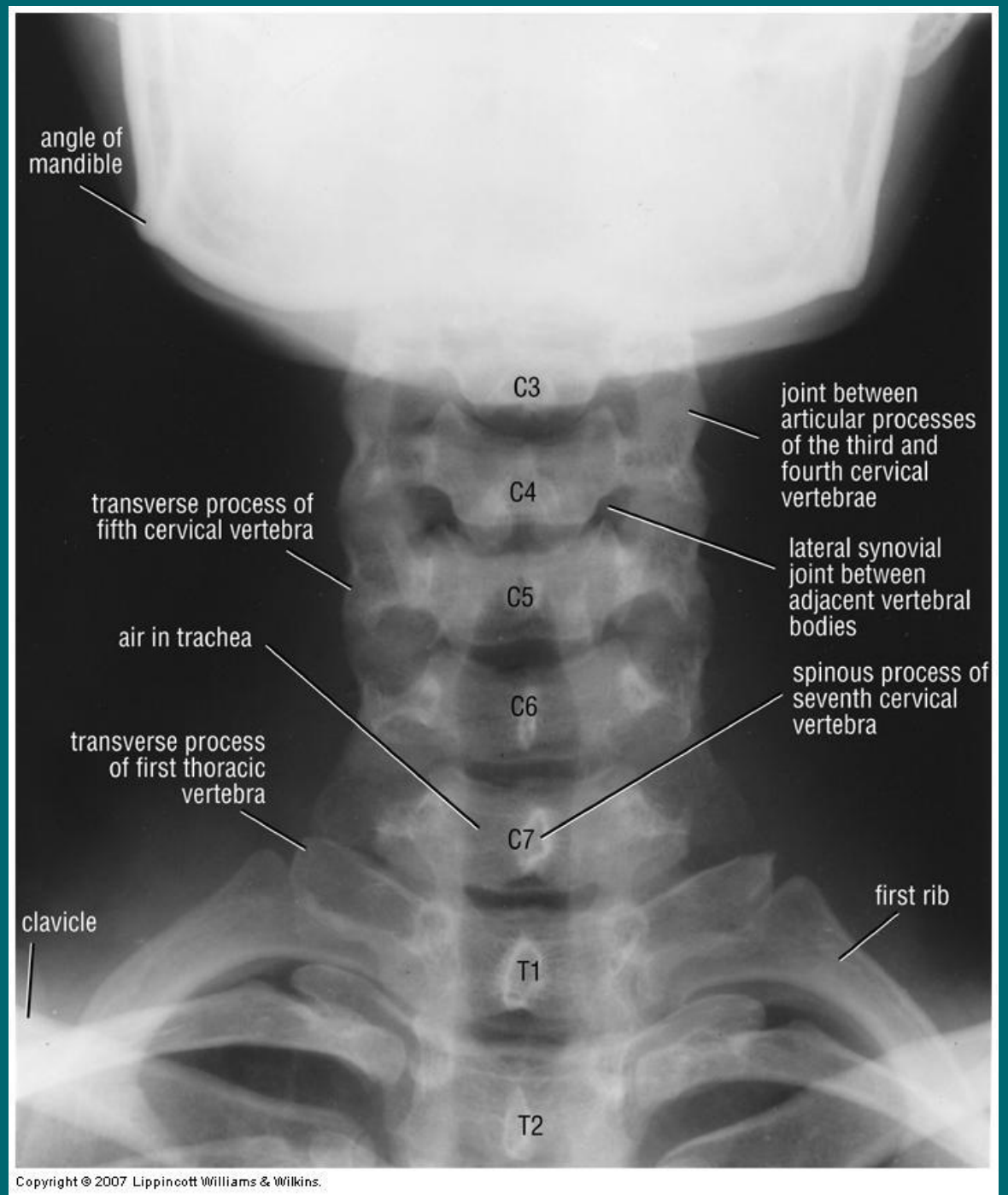
- Smaller bodies
  - Uncus of the body (elevated superolateral margin)
- Larger vertebral foramen
  - triangular in shape
- Transverse processes
  - shorter
  - transverse foramen
    - For vertebral artery and veins except C7
    - Anterior and posterior tubercles
      - For levator scapulae & scalenes
      - Carotid tubercle (anterior tubercle of C6)
- Spinous processes of C2 to C6 often bifid
  - C7 (prominens) has long prominent spinous process
- Superior articular processes face posteriosuperior




Left anterior superior oblique views of "typical" presacral vertebrae

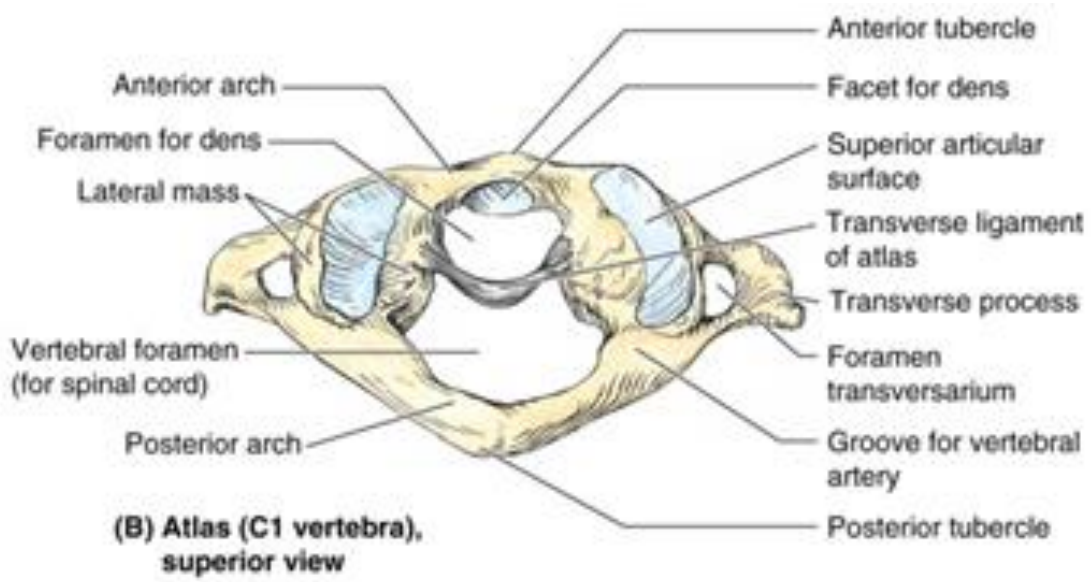


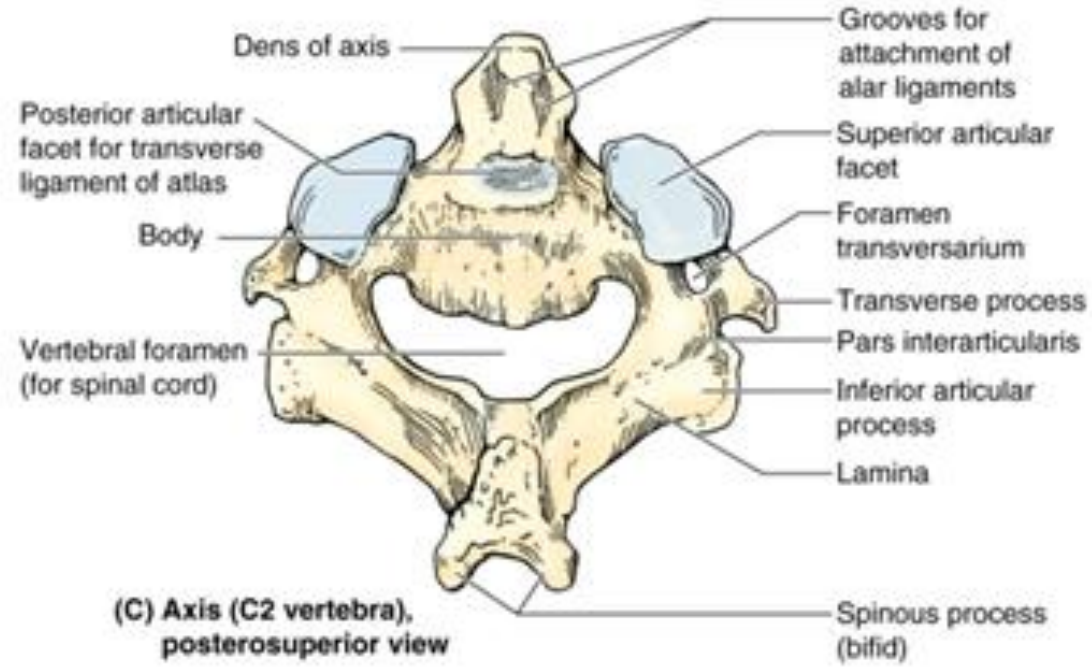
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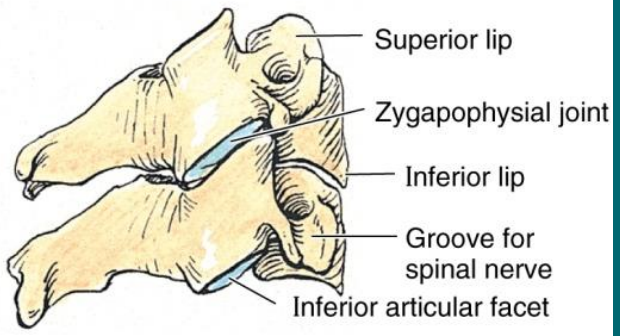
# Atypical Cervical Vertebrae

- C1 (Atlas)
  - NO body or spinous process
  - Ring shape bone
    - Anterior arch
      - Anterior tubercle
    - Posterior arch
      - Posterior tubercle
      - Groove for the vertebral a.
  - Lateral masses
    - Superior facets for occipital condyles
    - Transverse process
      - More laterally placed than others (atlas is the widest)
  - Nodding movement at atlanto-occipital joint signifies "yes"
- C2 (Axis)
  - Dens or odontoid process
    - from body of axis
  - Superior articular facets

Pivotal movement at atlanto-axial joint signifies "no" •

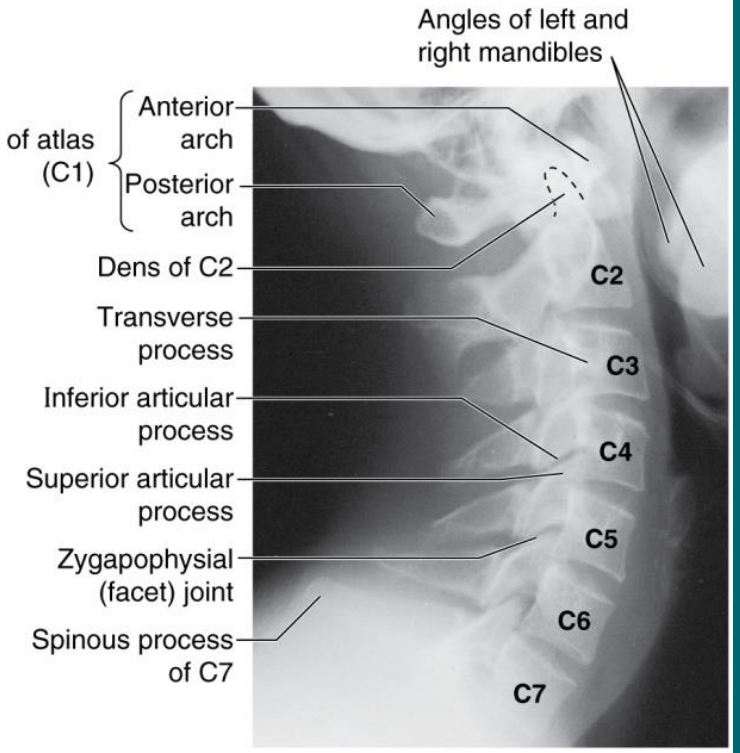





**(B) Lateral view, articulated typical cervical vertebrae C4 and C5**

Typical cervical vertebrae

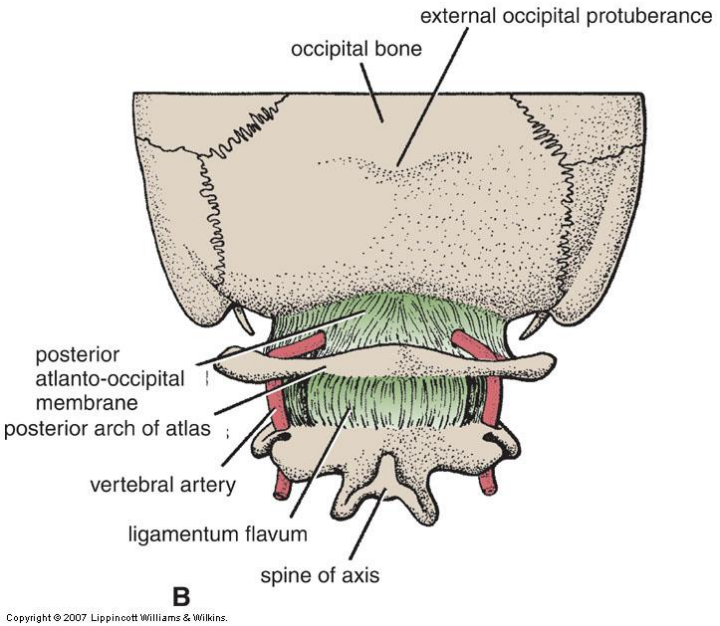


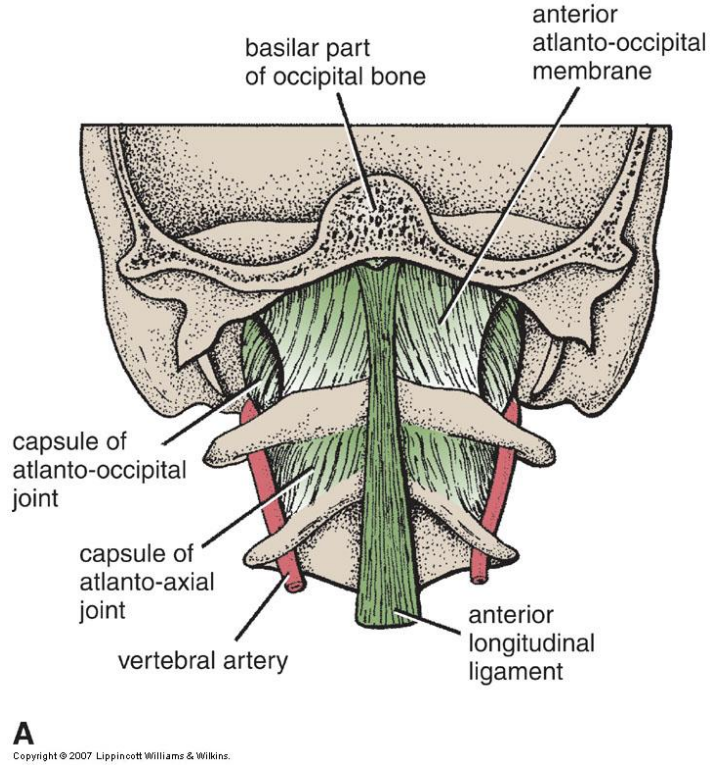

**(C) Lateral radiograph, vertebrae C1-C7**

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# Atlanto-Occipital Joints

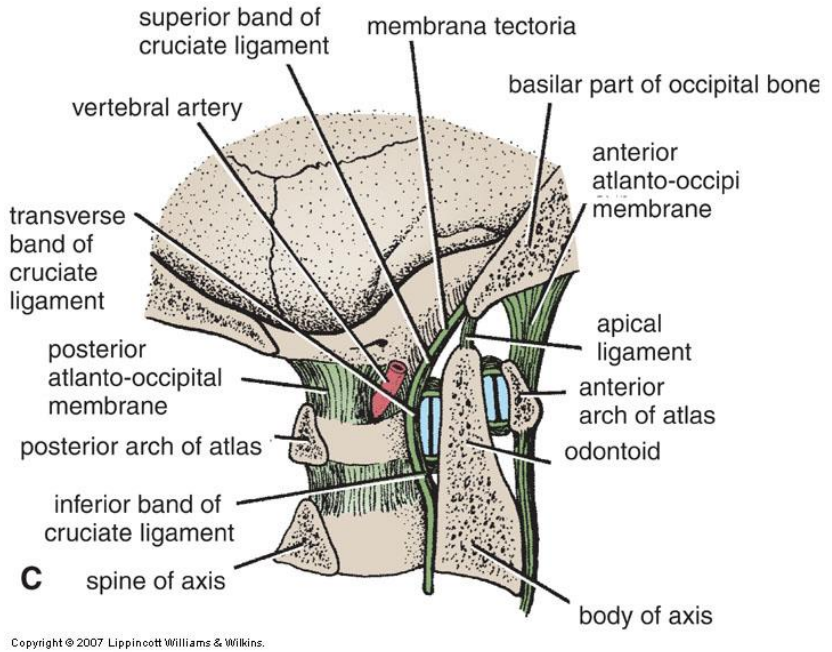
- Between occipital condyles and lateral masses of atlas
- Synovial joints
- Flexion & extension
- Ligaments
  - Anterior atlanto-occipital membrane
    - Continuation of anterior longi ligament
    - Between anterior arch of atla margin of foramen magnum
  - Posterior atlanto-occipital membrane
    - Connects posterior arch with margin of foramen magnum

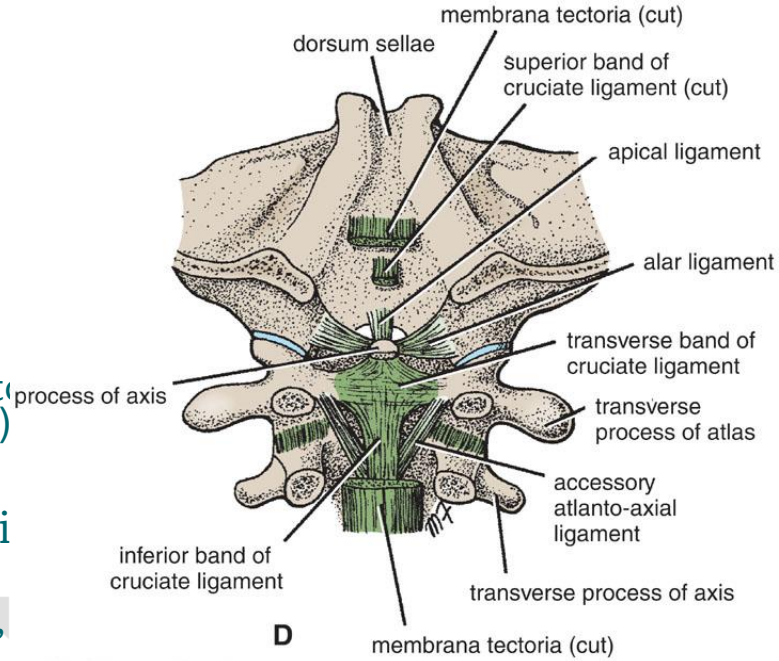


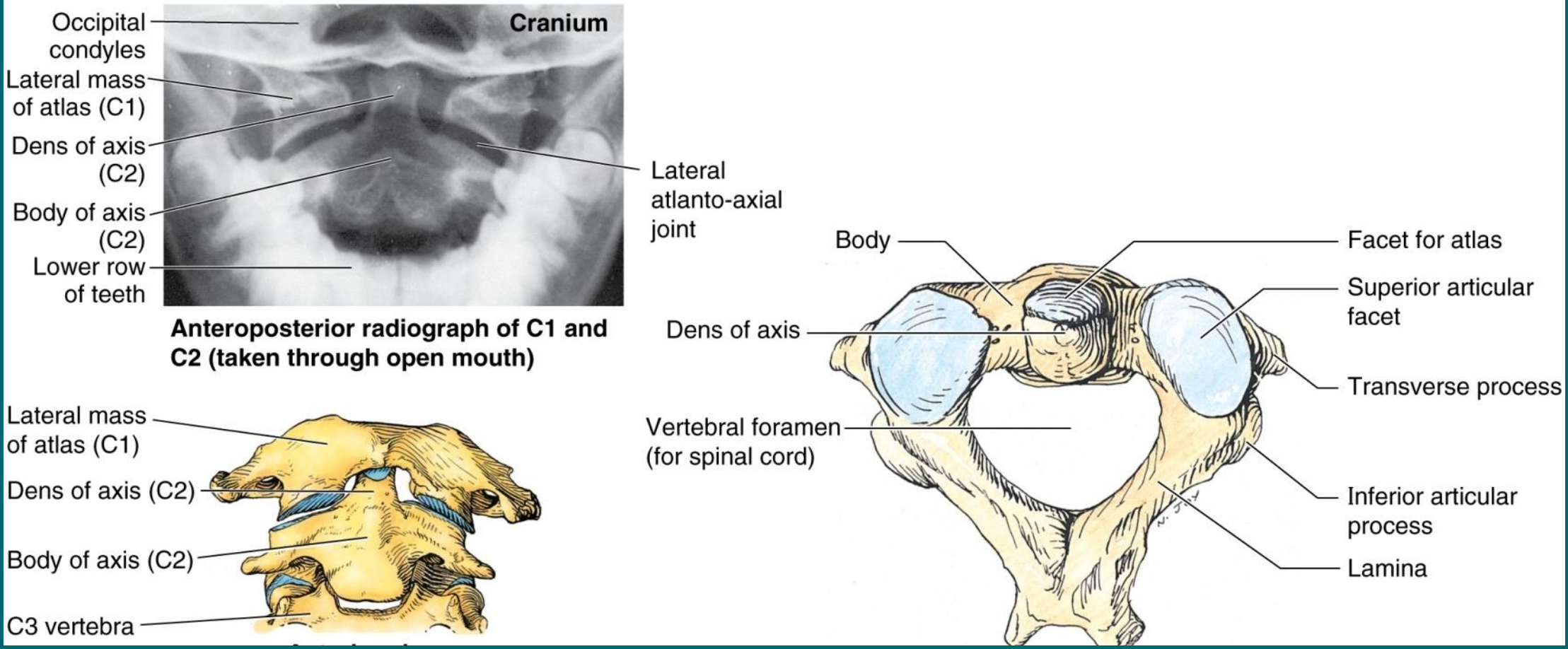



# Atlanto-Axial Joints

- Three synovial joints
  - Between the dens of axis & anterior arch
  - Between the lateral masses
- Rotation movement
- Ligaments
  - Apical ligament
    - Between apex of dens & margin of foramen magnum
  - Alar ligament
    - Lateral to the apical ligament
    - Between dens & occipital condyles
  - Cruciate ligament
    - Transverse band
      - Connects the lateral masses of atlas
    - Vertical band
      - Between the body of axis (posterior) t margin of foramen magnum (anterior)
  - Membrana tectoria
    - Continuation of the posterior longitudi ligament
    - Posterior to dens of axis & apical, alar, ligaments




Occipital condyles
Lateral mass of atlas (C1)
Dens of axis (C2)
Body of axis (C2)
Lower row of teeth
Cranium
Lateral atlanto-axial joint
Anteroposterior radiograph of C1 and C2 (taken through open mouth)
Lateral mass of atlas (C1)
Dens of axis (C2)
Body of axis (C2)
C3 vertebra
Body
Facet for atlas
Superior articular facet
Dens of axis
Transverse process
Vertebral foramen (for spinal cord)
Inferior articular process
Lamina

# Cervical Vertebral Muscles

- Suboccipital and deep neck muscles
  - ➢ Mainly extends head and neck
  - Rectus capitis posterior major m.
  - Rectus capitis posterior minor m.
  - Oblique capitis inferior m.
  - Oblique capitis superior m.

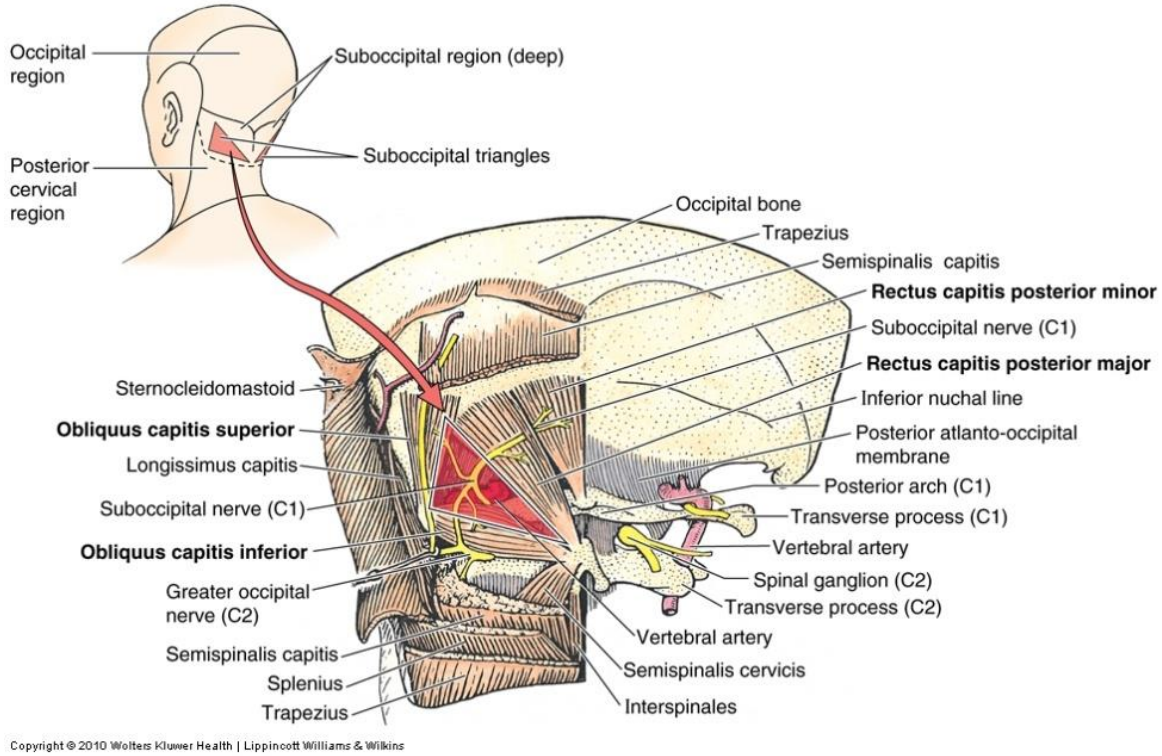
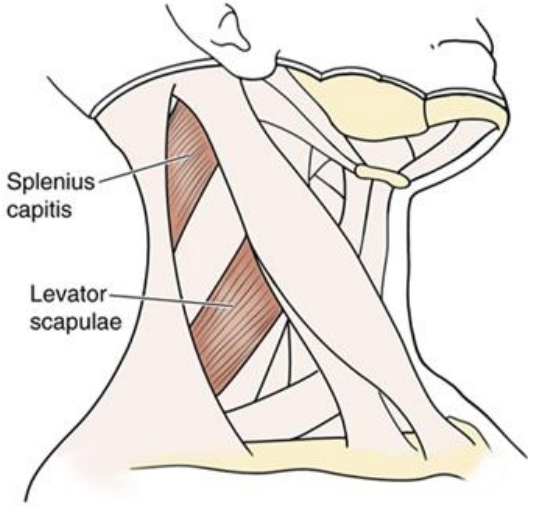



TABLE 4.9. SUBOCCIPITAL MUSCLES AND SUBOCCIPITAL TRIANGLE

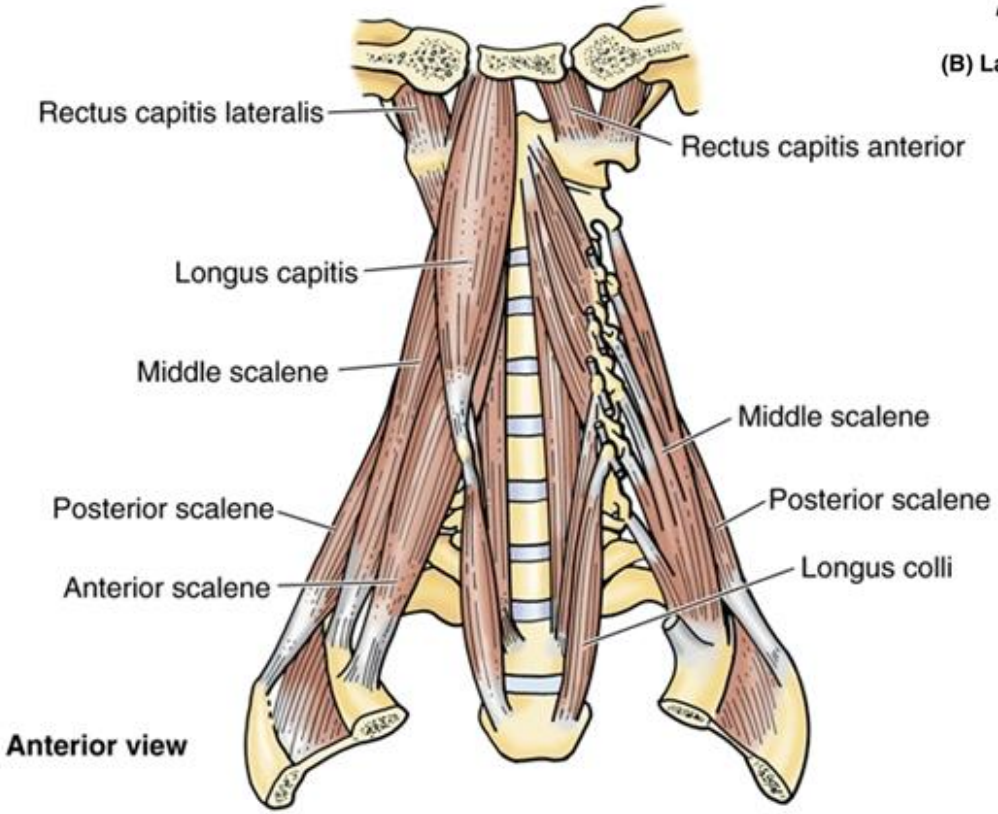
| Suboccipital Muscles | | |
|---|---|---|
| **Muscle** | **Origin** | **Insertion** |
| Rectus capitis posterior major | Spinous process of vertebra C2 | Lateral part of inferior nuchal line of occipital bone |
| Rectus capitis posterior minor | Posterior tubercle of posterior arch of vertebra C1 | Medial part of inferior nuchal line of occipital bone |
| Obliquus capitis inferior | Posterior tubercle of posterior arch of vertebra C2 | Transverse process of vertebra C1 |
| Obliquus capitis superior | Transverse process of vertebra C1 | Occipital bone between superior and inferior nuchal lines |

# Cervical Prevertebral Muscles

- Anterior vertebral muscles
  - ➢ Mainly flexes head and neck
  - Longus coli m.
  - Longus capitis m.
  - Rectus capitis anterior m.
  - Anterior scalene m.
- Lateral vertebral muscles
  - ➢ Mainly laterally flex head and neck
  - Rectus capitis lateralis m.
  - Splenius capitis m.
  - Levator scapulae m.
  - Middle scalene m.
  - Posterior scalene m.



(B) Lateral view



(A) Anterior view

**TABLE 8.4. PREVERTEBRAL MUSCLES**

| Muscle | Superior Attachment | Inferior Attachment | Innervation | Main Action(s) |
|---|---|---|---|---|
| Anterior vertebral muscles | | | | |
| Longus colli | Anterior tubercle of C1 vertebra (atlas); bodies of C1–C3 and transverse processes of C3–C6 vertebrae | Bodies of C5–T3 vertebrae; transverse processes of C3–C5 vertebrae | Anterior rami of C2–C6 spinal nerves | Flexes neck with rotation (torsion) to opposite side if acting unilaterally[a] |
| Longus capitis | Basilar part of occipital bone | Anterior tubercles of C3–C6 transverse processes | Anterior rami of C1–C3 spinal nerves | Flex head[b] |
| Rectus capitis anterior | Base of cranium, just anterior to occipital condyle | Anterior surface of lateral mass of atlas (C1 vertebra) | Branches from loop between C1 and C2 spinal nerves | |
| Anterior scalene | Transverse processes of C3–C6 vertebrae | 1st rib | Cervical spinal nerves C4–C6 | |

**TABLE 8.4. PREVERTEBRAL MUSCLES (Continued)**

| Muscle | Superior Attachment | Inferior Attachment | Innervation | Main Action(s) |
|---|---|---|---|---|
| Lateral vertebral muscles | | | | |
| Rectus capitis lateralis | Jugular process of occipital bone | Transverse process of atlas (C1 vertebra) | Branches from loop between C1 and C2 spinal nerves | Flexes head and helps stabilize it[b] |
| Splenius capitis | Inferior half of nuchal ligament and spinous processes of superior six thoracic vertebrae | Lateral aspect of mastoid process and lateral third of superior nuchal line | Posterior rami of middle cervical spinal nerves | Laterally flexes and rotates head and neck to same side; acting bilaterally, extends head and neck[c] |
| Levator scapulae | Posterior tubercles of transverse processes C2–C6 vertebrae | Superior part of medial border of scapula | Dorsal scapular nerve C5 and cervical spinal nerves C3 and C4 | Downward rotation of scapula and tilts its glenoid cavity inferiorly by rotating scapula |
| Middle scalene | Posterior tubercles of transverse processes of C5–C7 vertebrae | Superior surface of 1st rib; posterior to groove for subclavian artery | Anterior rami of cervical spinal nerves | Flexes neck laterally; elevates 1st rib during forced inspiration[a] |
| Posterior scalene | | External border of 2nd rib | Anterior rami of cervical spinal nerves C7 and C8 | Flexes neck laterally; elevates 2nd rib during forced inspiration[a] |

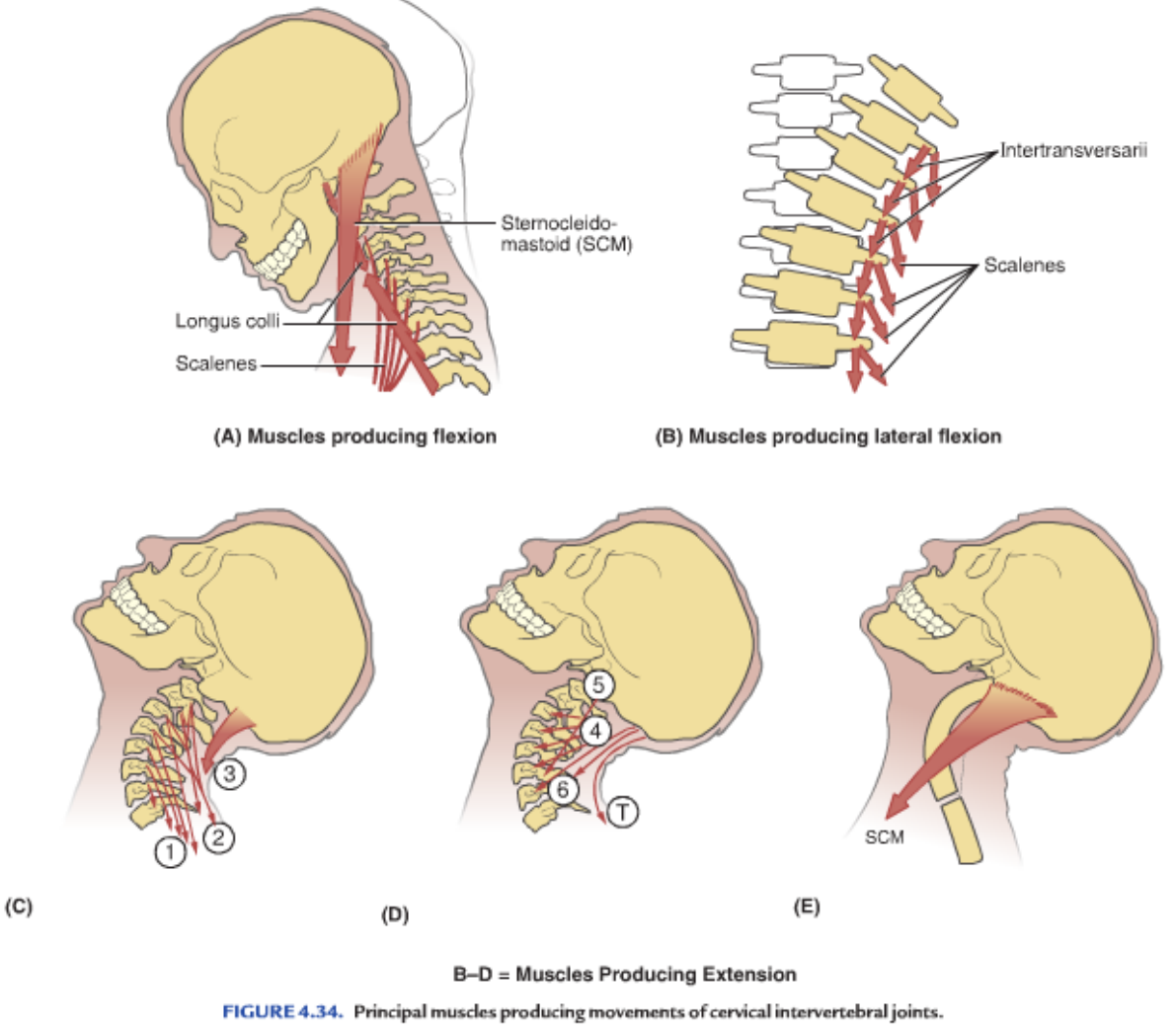
[a]Flexion of neck = anterior (or lateral) bending of cervical vertebrae C2–C7.

[b]Flexion of head = anterior (or lateral) bending of the head relative to the vertebral column at the atlanto-occipital joints.

[c]Rotation of the head occurs at the atlanto-axial joints.

# Movements of the Cervical Vertebral Column

- Flexion
  - Longus cervicis, scalenus anterior & SCM
- Extension
  - Postvertebral mm.
- Lateral flexion
  - Scalenus anterior& medius, trapezius & SCM
- Rotation
  - SCM on one side & splenius on the other side




(A) Muscles producing flexion (B) Muscles producing lateral flexion

(C) (D) (E)

B–D = Muscles Producing Extension

FIGURE 4.34. Principal muscles producing movements of cervical intervertebral joints.

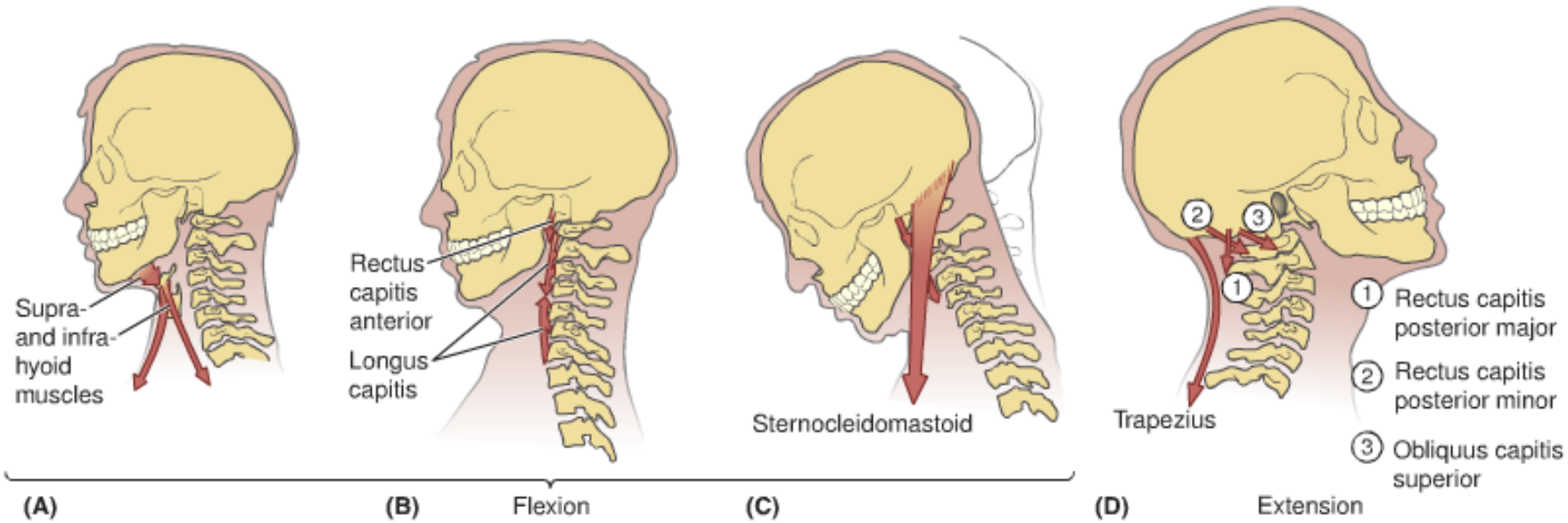
TABLE 4.7. PRINCIPAL MUSCLES PRODUCING MOVEMENT OF CERVICAL INTERVERTEBRAL JOINTS

| Flexion | Extension | Lateral Bending | Rotation (not shown) |
|---|---|---|---|
| Bilateral action of<br>Longus coli<br>Scalene<br>Sternocleidomastoid | Deep neck muscles<br>*1*, semispinalis cervicis and iliocostalis cervicis<br>*2*, splenius cervicis and levator scapulae<br>*3*, splenius capitis<br>*4*, multifidus<br>*5*, longissimus capitis<br>*6*, semispinalis capitis<br>*T*, trapezius | Unilateral action of<br>Iliocostalis cervicis<br>Longissimus capitis and cervicis<br>Splenius capitis and cervicis<br>Intertransversarii and scalenes | Unilateral action of<br>Rotatores<br>Semispinalis capitis and cervicis<br>Multifidus<br>Splenius cervicis |

# Movements of the Cervical Vertebral Column

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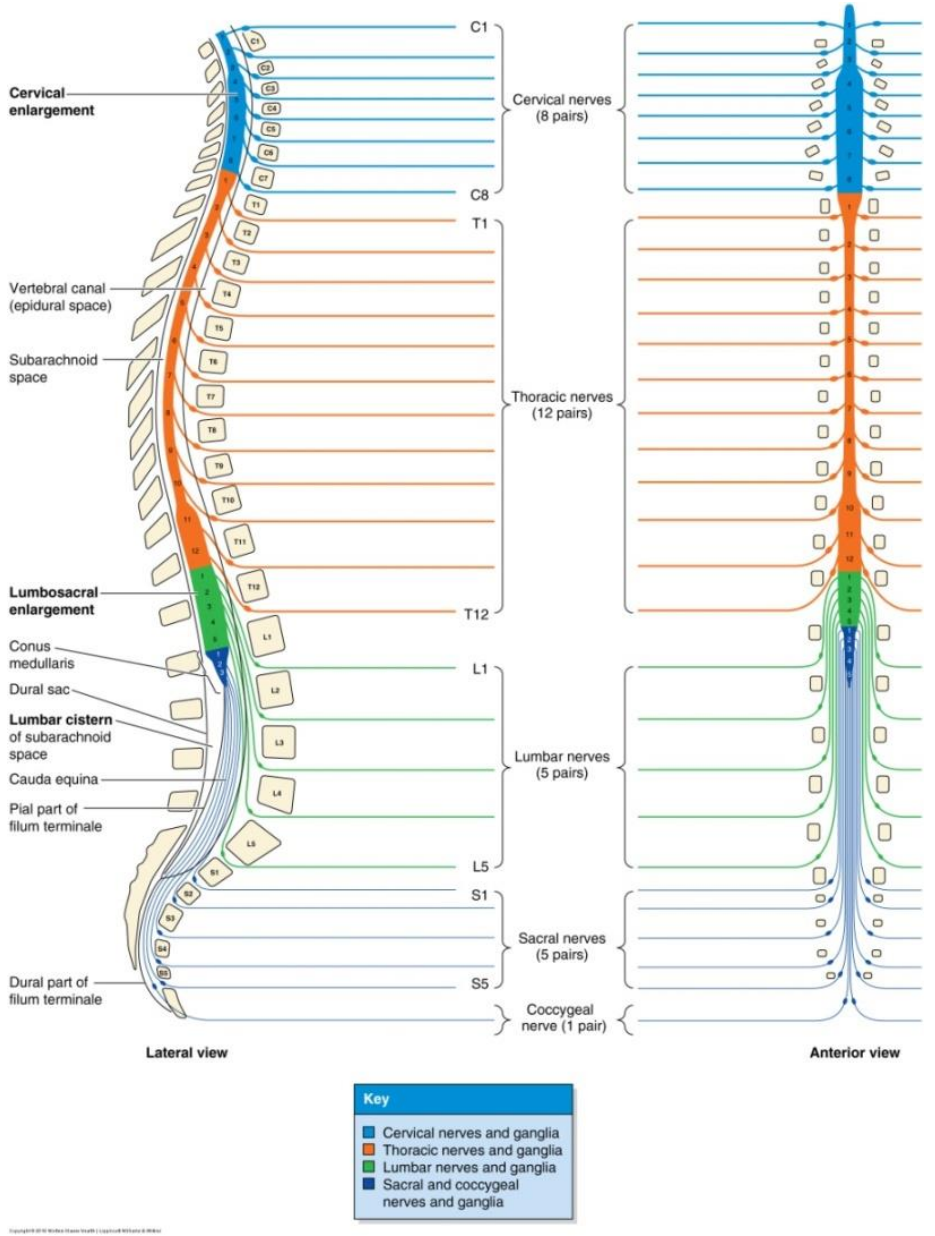
**TABLE 4.10. PRINCIPAL MUSCLES PRODUCING MOVEMENT OF ATLANTO-OCCIPITAL JOINTS**




| Flexion | Extension | Lateral Flexion (not shown) |
|---|---|---|
| Longus capitis | Rectus capitis posterior, major and minor | Sternocleidomastoid |
| Rectus capitis anterior | Obliquus capitis superior | Obliquus capitis superior |
| Anterior fibers of sternocleidomastoid | Splenius capitis | Rectus capitis lateralis |
| Suprahyoid and infrahyoid muscles | Longissimus capitis | Longissimus capitis |
| | Trapezius | Splenius capitis |

# Cervical Vertebral Canal: Content

- Meninges
  - Dura matter
    - Continuous with cranial dura matter (meningeal layer)
  - Arachnoid matter
  - Pia matter
- Lower part of medulla oblongata
- Cervical segments of the spinal cord
  - Contain the upper motor neurons for the upper and lower limbs
  - Other descending fibers to the spinal cord (e.g. reticulospianl fibers)
  - Contain the ascending (sensory) fibers from the neck below
  - Cervical enlargement
    - Innervation for the upper limb
      - Lower motor neurons
- Cervical spinal nerves
  - C1-C8
    - C1-C7 exit above the corresponding vertebra
      - C1 exit between the atlas and the occipital bone
    - C8 exit between the C7 and T1 vertebrae
  - C1-C4 form the cervical plexus
  - C5-T1 form the brachial plexus

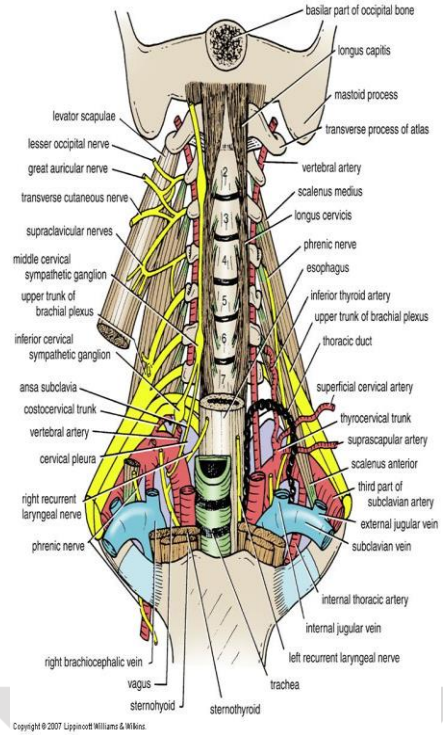


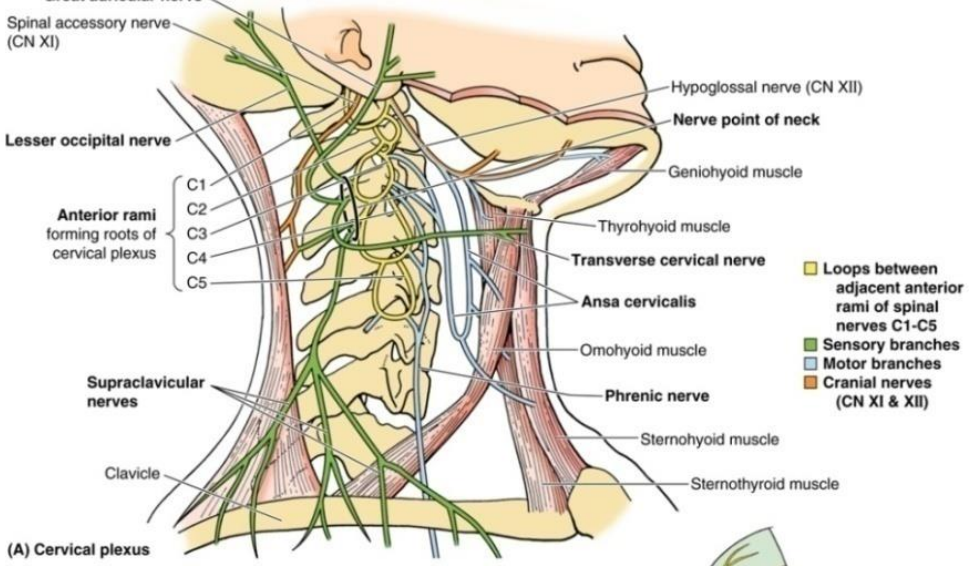
# Impingement on the Cervical Spinal Cord

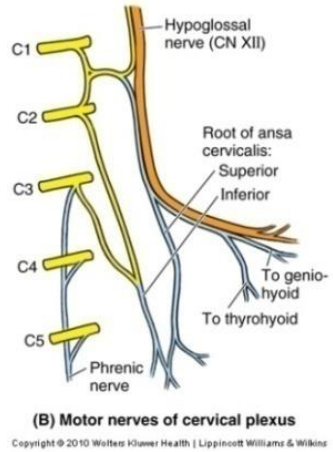
- Causes
  - Intervertebral disc pathology
    - Inflammation
    - Degeneration
    - Herniation
  - Traumatic injuries to the neck
  - Tumors, infections, congenital anomalies to the vertebral column
- Effects of impingement on the cervical spinal cord
  - ➢Depend on the level of impingement
  - Respiratory dysfunction
  - Sensory dysfunctions of the neck, upper & lower limbs (pain, numbness)
  - Motor dysfunctions of the neck, upper & lower limbs (stiffness, weakness, paralysis)
  - Visceral organs dysfunction (urination, defecation, erection, ejaculation)

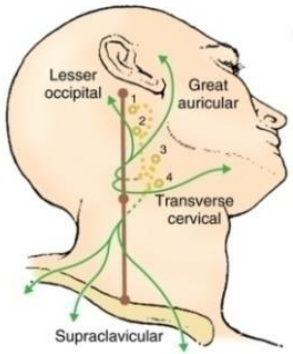
# Cervical Plexus

- Ventral rami C1-C4
- Relations
  - Anterior to levator scapulae m. and middle scalene m.
  - Posterior to the sternocleidomastoid m.
  - Subcutaneous branches emerge behind the lateral border of the sternocleidomastoid m.
- Branches
  - Cutaneous branches
    - Lesser occipital n. (C2)
    - Greater auricular n. (C2-C3)
    - Transverse cervical n. (C2-C3)
    - Supraclavicular nn. (C3-C4)
  - Muscular branches
    - Ansa cervicalis (Infrahyoid mm.)
      - Descending branch from hypoglossal n. (C1)
      - Descending cervical n. (C2-C3)
    - Phrenic n. (C3-C5)
      - Diaphragm



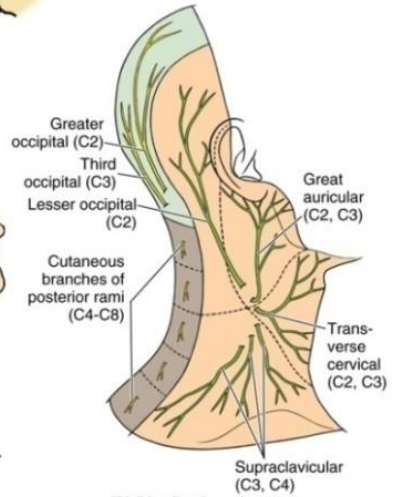



**Table 17-1** Summary of the Branches of the Cervical Plexus and Their Distribution

| Branches | Distribution |
|---|---|
| Cutaneous | |
| Lesser occipital | Skin of scalp behind ear |
| Greater auricular | Skin of parotid salivary gland, auricle, and angle of jaw |
| Transverse cutaneous | Skin over side and front of neck |
| Supraclavicular | Skin over upper part of chest and shoulder |
| Muscular | |
| Segmental | Prevertebral muscles, levator scapulae |
| Ansa cervicalis (C1, 2, 3) | Omohyoid, sternohyoid, sternothyroid |
| C1 fibers via hypoglossal nerve | Thyrohyoid, geniohyoid |
| Phrenic nerve (C3, 4, 5) | Diaphragm (most important muscle of respiration) |
| Sensory | |
| Phrenic nerve (C3, 4, 5) | Pericardium, mediastinal parietal pleura, and pleura and peritoneum covering central diaphragm |

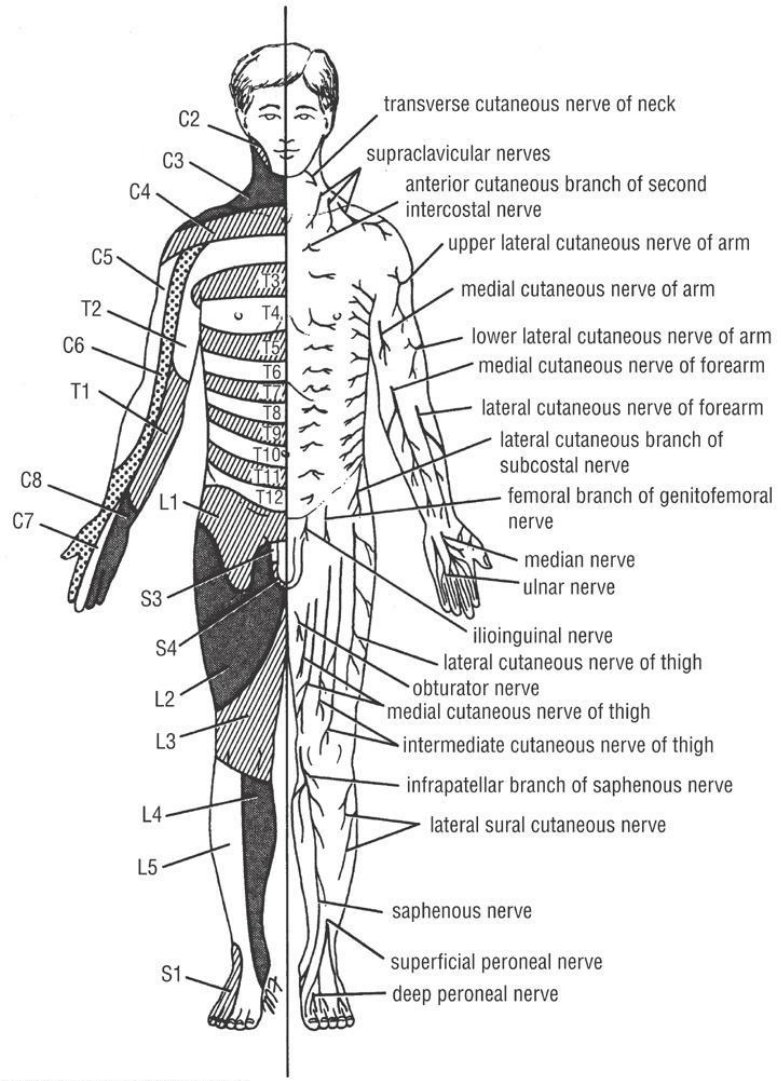
# Intervertebral Foramen Encroachment

- Causes
  - IV disc herniation
  - IV disc degeneration
  - Bone spurs
- Effects (depend on the spinal nerves involved)
  - Numbness
  - Pain (local or along the affected nerve)
  - Weakness

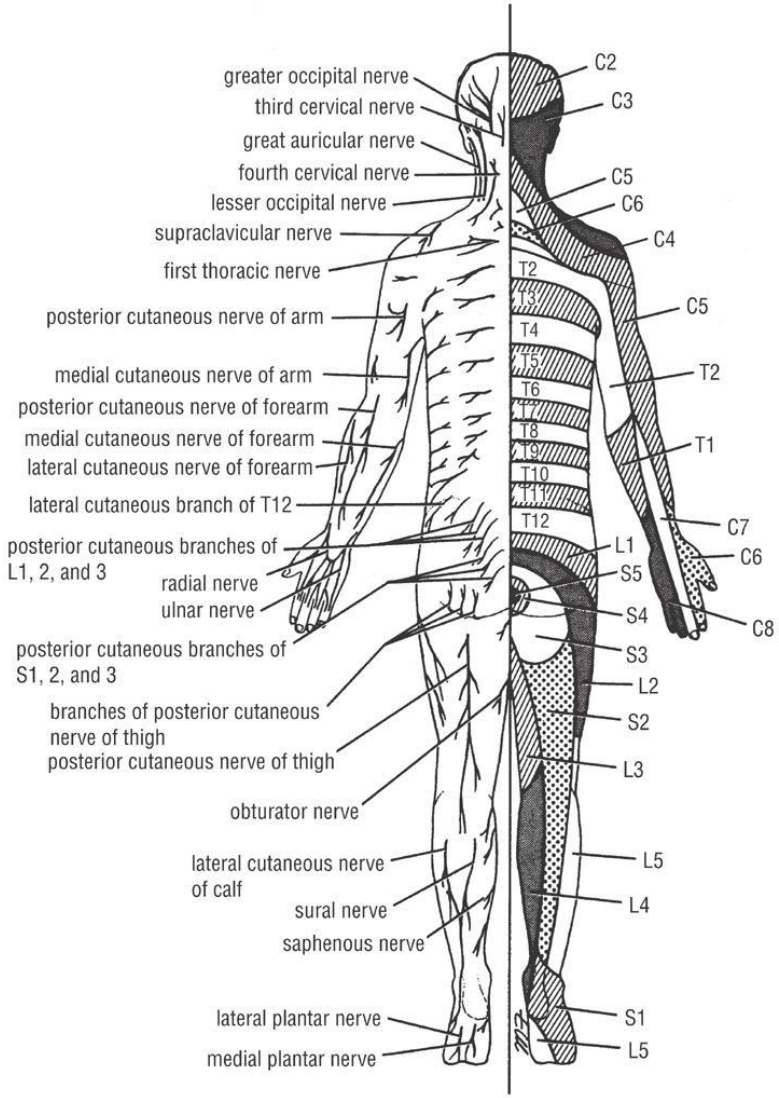
# The specific neurovascular manifestations of acute cervical disc herniation

| IV disc level | Nerve root level | Manifestations | Reflexes |
|---|---|---|---|
| C2 | C3 | posterior neck numbness and pain radiating to the mastoid and ear | reflexes test normal |
| C3 | C4 | posterior neck numbness and pain radiating along the levator scapulae muscle and sometimes to the pectoralis | reflexes test normal |
| C4 | C5 | lateral neck, shoulder, and arm pain and paresthesia, deltoid weakness and possible atrophy, hypesthesia of C5 root distribution over middle deltoid area (axillary nerve distribution). | reflexes test normal |
| C5 | C6 | pain radiating down the lateral arm and forearm into the thumb and index finger, hypesthesia of the lateral forearm and thumb | decreased biceps reflex, biceps and supinator weakness |
| C6 | C7 | pain radiating down the middle forearm to the middle fingers, hypesthesia of the middle fingers | decreased triceps and radial reflexes, triceps and grip weakness |
| C7 | C8 | possible pain radiating down the medial forearm and hand, ulnar hypesthesia, intrinsic muscle weakness of the hand. However, these symptoms are uncommon | reflexes test normal |

# Dermatomes & Cutaneous Nerves of the Upper Limb




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# Cutaneous Innervation of the Upper Limb

Musculoskeletal

Axillary

Radial

Ulnar

Median

Medial cord

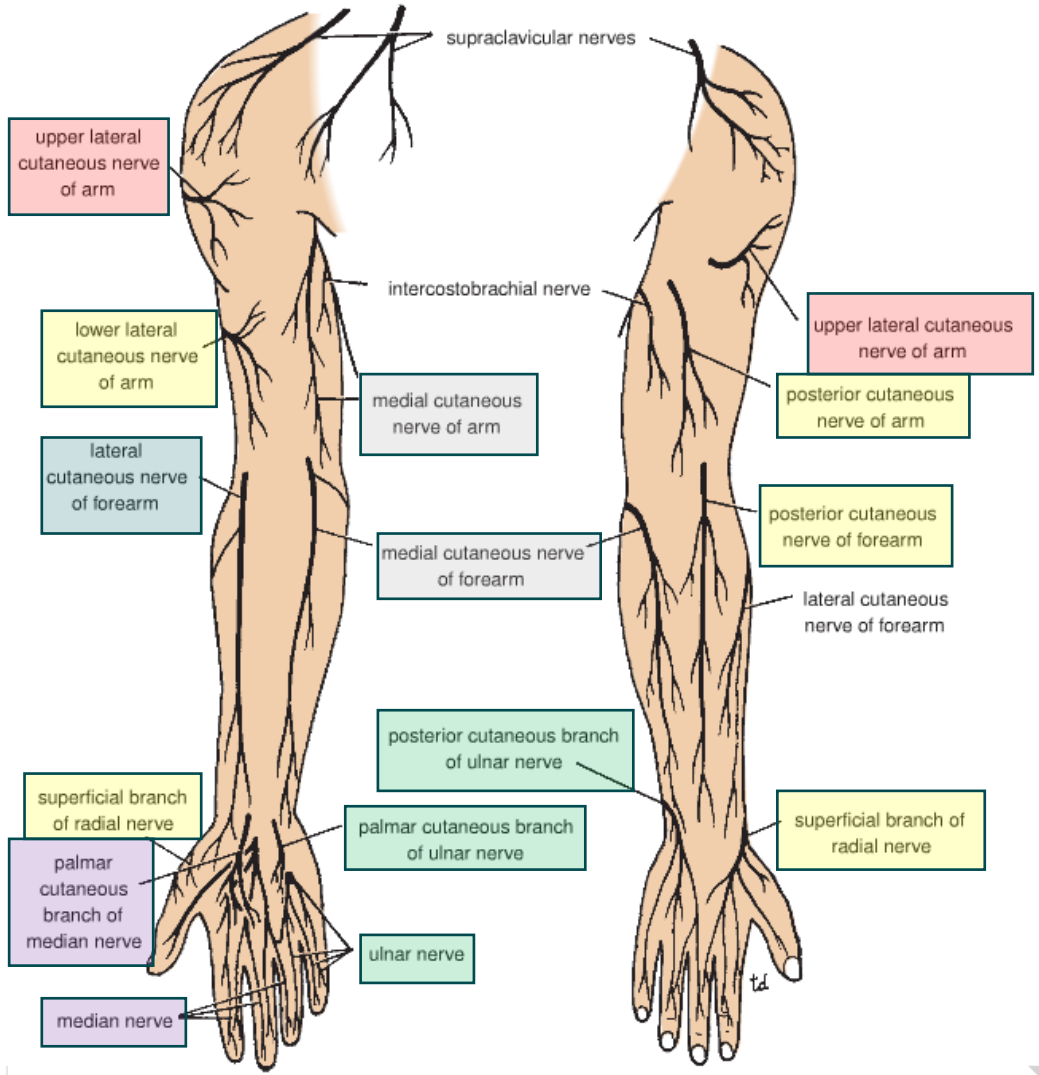
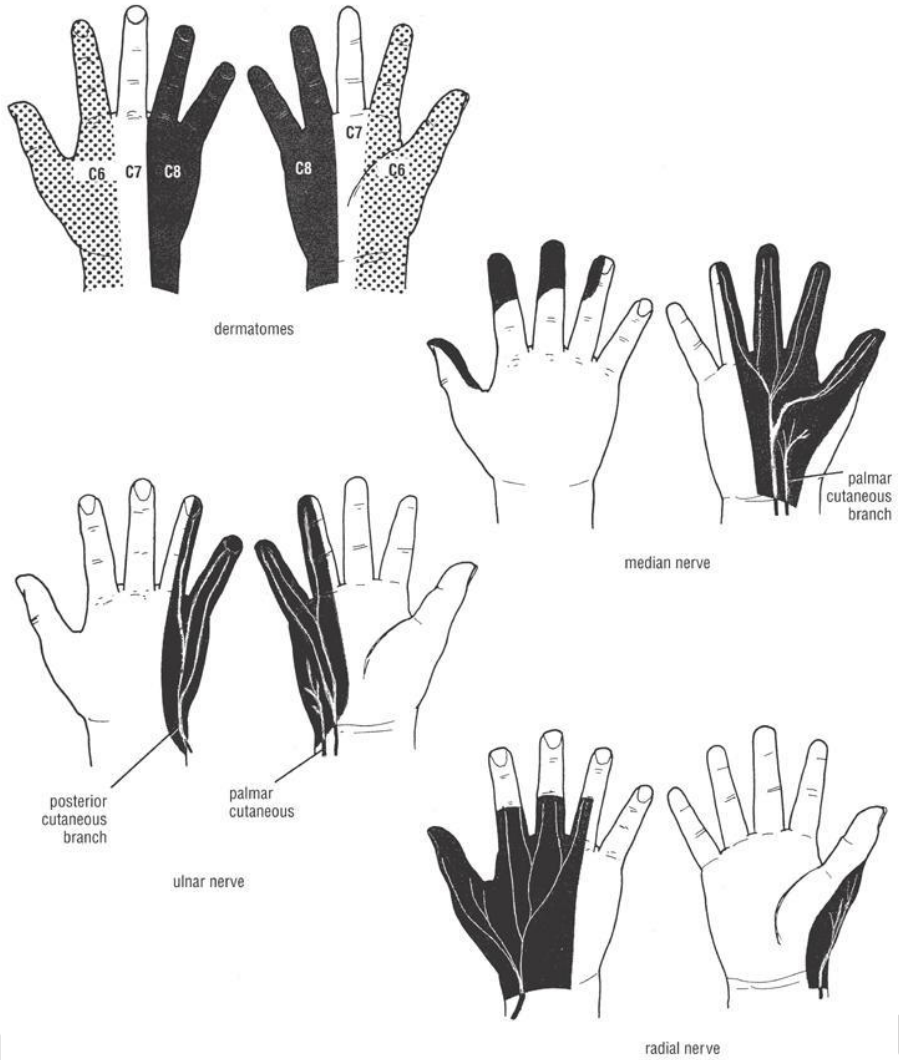



Figure 17-8 Cutaneous innervation of the upper limb.

# Dermatomes & Cutaneous Nerves of the Hand




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# Dermatome Tests for the Upper Limb

**TABLE 6.1. DERMATOMES OF UPPER LIMB**

| Spinal Segment/Nerve(s) | Description of Dermatome(s) |
|---|---|
| C3, C4 | Region at base of neck, extending laterally over shoulder |
| C5 | Lateral aspect of arm (i.e., superior aspect of abducted arm) |
| C6 | Lateral forearm and thumb |
| C7 | Middle and ring fingers (or middle three fingers) and center of posterior aspect of forearm |
| C8 | Little finger, medial side of hand and forearm (i.e., inferior aspect of abducted arm) |
| T1 | Medial aspect of forearm and inferior arm |
| T2 | Medial aspect of superior arm and skin of axilla[a] |

[a]Not indicated on the Keegan and Garrett (1948) dermatome map. However, pain experienced during a heart attack, considered to be mediated by T1 and T2, is commonly described as "radiating down the medial side of the left arm."

# Examination of Different Sensory Modalities

- Light touch
  - light touch of a finger or a piece of cotton
- Sharp touch (pinprick)
  - Test using a dedicated disposable pin
- Vibration sense
  - Use a 128 Hz tuning fork
  - Begin with one of the distal interphalangeal joints
  - If no vibration is sensed, move backwards
- Joint position sense (proprioception)
  - Begin with one of the distal interphalangeal joints
  - Move the distal phalanx up and down randomly. patient should know the direction of movement
  - If patient did not know, move backwards

➢ Tests are done while patient is closing eyes

➢ Both sides should be tested and compared











# Myotome Tests for the Upper Limb

- Abduction of Arm ($C_5$)
  - Supraspinatus –Supraspinatus Nerve
  - Deltoid-Axillary Nerve
- Arm Adduction ($C_7$)
  - Pectoralis Major- Pectoral Nerves
  - Latissimus Dorsi-Thoracodorsal Nerve
- Forearm Flexion ($C_{5-6}$)
  - Brachialis-Musculocutaneous
  - Biceps Brachii-Musculocutaneous







# Myotome Tests for the Upper Limb

- Forearm Extension ($C_7$)
  - Triceps Brachii-Radial nerve
- Wrist Flexion ($C_{7,8}$ $T_1$)
  - Flexor Carpi Radialis-Median Nerve
  - Flexor Carpi Ulnaris-Ulnar Nerve
- Wrist Extension ($C_{7,8}$)
  - Extensor Carpi Radialis (Longus and Brevis)-Radial Nerve
  - Extensor Carpi Ulnaris-Posterior Interosseous Nerve







# Myotome Tests for the Upper Limb

- Finger Flexion ($C_{7,8}T_1$)
  - Flexor Digitorum Superficialis-Median Nerve
  - Flexor Digitorum Profudus-Ulnar & Anterior Interosseous branch of Median
- Finger Extension ($C_7$)
  - Extensor Digitorum-Posterior Interosseous branch of Radial
- Thumb Abduction ($C_{7,8}$ $T_1$)
  - Abductor Pollicis Longus-Radial Nerve
  - Abductor Pollicis Brevis-Median Nerve







# Myotome Tests for the Upper Limb

- Thumb Opposition ($T_1$)
  - Opponens Pollicis: Median Nerve
- Finger Abduction ($C_8, T_1$)
  - Dorsal Interossei & Abductor Digiti Minimi-Ulnar Nerve
- Finger Adduction ($C_8, T_1$)
  - Palmar interossei-Ulnar Nerve
  - http://www.youtube.com/watch?v=rKiTwagLYck







# Tendon Reflexes & Segmental Innervation of the Upper Limb Muscles

- Biceps brachii tendon reflex
  - C5 &C6
  - Flexion of the elbow joint
  - Tapping on the biceps tendon
- Triceps tendon reflex
  - C6-C8
  - Extension of the elbow joint
  - Tapping on the triceps tendon
- Brachioradialis tendon reflex
  - C5-C7
  - Supination of the radioulnar joint
  - Tapping the brachioradialis tendon

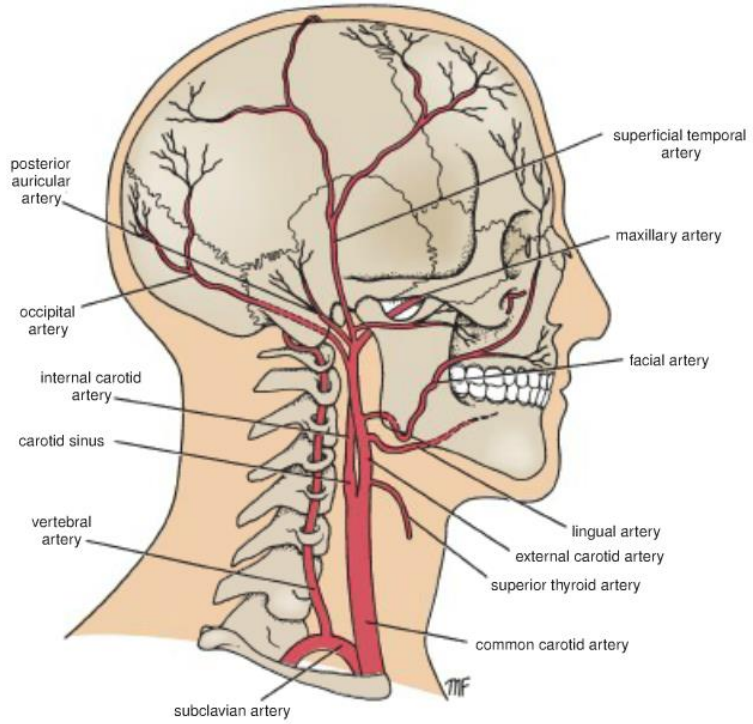


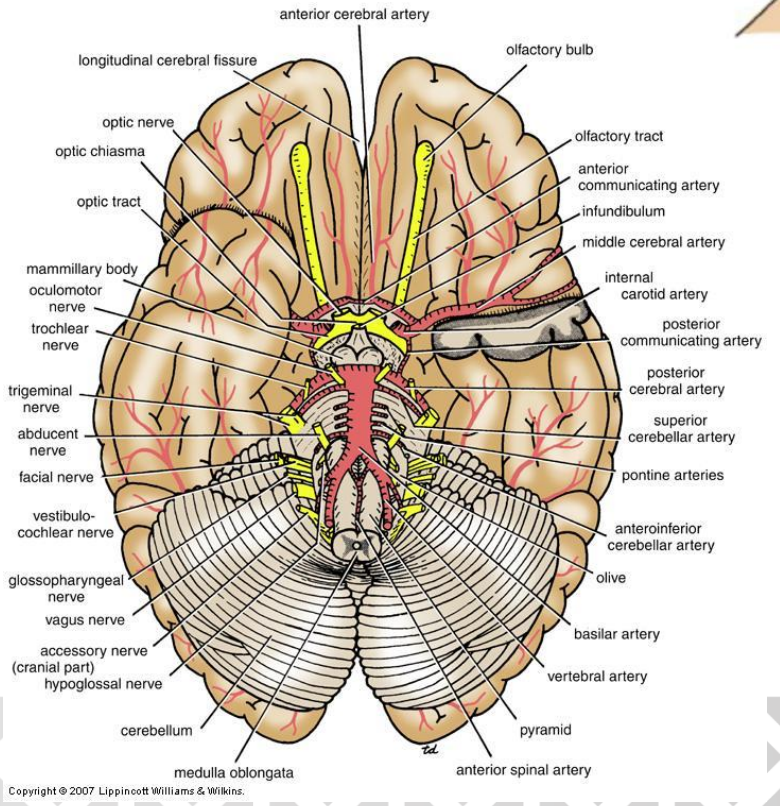




# Vertebral Artery

- Branch of the 1st part of the subclavian a.
- Ascends the neck through the upper sex transverse vertebrae
- Inters the skull through foramen magnum
- Both sides unit to form the basilar a.
  - Ascend in front of pons
  - Participate in forming circle of Willis
  - Branches
    - Anterior inferior cerebellar aa.
    - Pontine aa.
    - Superior cerebellar aa.
    - Posterior cerebral aa. (end branches)
      - Inferior & posterior parts of cerebrum
- Branches
  - Meningeal aa.
  - Anterior spinal a.
  - Posterior spinal aa.
  - Posterior inferior cerebellar a.
  - Medullary aa.



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# Thanks